LIVE ENGINEERING VALIDATION & ARCHITECTURE

# SLEEX & SLEEX-VPS

## Technical Architecture, Multi-Cloud Ephemeral Mesh (AWS, GCP, Azure) & Live Testing Phase.

An edge-native, mobile-first autonomous software engineering studio. Live dogfooding verification across Cloudflare Workers SQLite Durable Objects, OpenClaw 2026.11 multi-agent coordination, and elastic cloud workers.

PREPARED BY
Engineering Architecture & Quality Assurance

PRIMARY OBJECTIVE
Live Dogfooding, Multi-Cloud Verification & System Reliability

CODE REPOSITORIES
pixca-net/sleex & pixca-net/sleex-vps

PRODUCTION APP
https://sleex.org

# Live Testing Phase: Objectives, Scope & Readiness

Validating end-to-end autonomous engineering execution and multi-cloud worker reliability under real-world loads.

**-92%**
OPERATING COST
vs. traditional cloud IDEs & heavy VM sandboxes

**24 / 7**
AUTONOMOUS LOOP
Continuous triage, coding, review, and merging

**3 Clouds**
WORKER MESH
AWS Lambda, GCP Cloud Run, Azure Container Apps

**<45s**
BUILD & TEST LOOP
Automated validation with zero manual file edits

## 🎯 Live Dogfooding Scope

Modern autonomous engineering requires thorough, transparent validation. In this live testing phase, Sleex validates real-world issue resolution, subagent coordination stability, and zero-leak credential isolation across active production checkouts.

- **Continuous Dogfooding:** Claw Team resolves live issues directly on `pixca-net/sleex` and `sleex-vps`.
- **Multi-Cloud Elasticity:** Live failover testing across AWS Lambda, GCP Cloud Run, and newly added Azure Container Apps Jobs.
- **Unit Economics Benchmark:** Real-world PR COGS measured at ~$0.18 per verified pull request.

## 🧪 Live Verification Criteria

The testing phase ensures full runtime stability, zero regression risk, and robust self-healing mechanics across all edge, VPS, and cloud worker tiers.

- **Automated Test Suites:** Comprehensive unit, integration, and profile drift tests covering all services.
- **Self-Healing Rollback:** 3-minute automatic blue/green rollback verified via `updater.mjs`.
- **Model Policy Enforcement:** Dynamic `goLimits` price bands and model fallback validation.

# The Problem: AI Coding is Broken, Expensive & Trapped

Why existing coding agents fail enterprise budgets and modern developer workflows.

### 💸 Crushing Token & Infra Costs

Agents dump entire codebases and mega-byte tool outputs into LLM context windows without compression. Teams burn $50–$150/day per active developer in raw model tokens and idle VM instances.

- No intelligent token caching or tool deduplication
- Always-on heavy cloud VMs billing 24/7
- Opaque vendor pricing markups

### 💻 Desktop & Laptop Confinement

Existing tools (Cursor, Windsurf, VS Code plugins) require the developer to sit physically in front of a multi-core machine. You cannot triage, review, or steer automated builds from your mobile device.

- Heavy local resource drain on laptops
- Zero native mobile steering experience
- Tasks terminate when laptop closes

### 🔒 Walled Garden Lock-In

OpenAI Codex locks you to GPT models; Claude Coworker restricts to Anthropic. When OpenAI or Anthropic suffers an API outage, enterprise engineering velocity grinds to a dead halt.

- Zero failover between model providers
- Proprietary black-box sandbox execution
- Data privacy and compliance concerns

### 🛑 Human Bottleneck & PR Bloat

AI generates code faster than engineers can review. Without automated multi-agent peer review, test verification, and automated PR merging, engineers spend 40% of their day manually auditing mediocre AI diffs.

- No independent reviewer agent gate
- Lack of automated browser visual QA
- Backlog of unmerged pull requests

# The Solution: The Sleex Autonomous Studio

A harmonized ecosystem of mobile-first edge UI, multi-agent brain, and ephemeral compute.

### 1. Ubiquitous Edge PWA

Zero-install Progressive Web App running on sleex.org. Instant Google OAuth, touch-optimized syntax highlighting, voice input, and live streaming transcripts. Monitor and steer tasks from anywhere.

`Cloudflare Workers + SQLite Durable Objects`

### 2. OpenClaw Autonomous Brain

Dedicated 6-agent autonomous workforce ("Claw Team"). Long-term cross-session memory, automated workboard progression, scheduled batch PR merges, and proactive daily digests.

`Coordinator • Coder • Reviewer • Maintainer • Designer`

### 3. Hybrid Ephemeral Sandboxes

Intelligent workload placement leasing AWS Lambda micro-workers, GCP Cloud Run, or Oracle ARM VPS via encrypted reverse SSH tunnels. Zero open public ports; near-zero idle cost.

`Dynamic Free-Tier Auto-Placement & Reverse SSH Gateway`

**Core Engineering Philosophy: "Less DIY, Leverage the Best"**

Sleex combines proven open standards: OpenCode SDK 2.2, OpenClaw 2026.11, Headroom token compression, and Cloudflare edge primitives to avoid reinventing commoditized agent plumbing.

PRODUCTION PROVEN

# Mission & Vision: Sovereign AI Engineering

Empowering every developer and company with an indefatigable, private software team.

OUR MISSION

## "To democratize autonomous software engineering by delivering a hyper-efficient, model-agnostic, mobile-accessible AI development team that runs 24/7 at near-zero idle cost."

We believe software development should not be bottlenecked by expensive SaaS tollbooths or fragile single-model monopolies. Sleex makes autonomous coding ubiquitous, transparent, and radically affordable.

OUR VISION

## "The sovereign AI operating system for modern software development—where every creator commands a self-healing, continuous delivery workforce with zero vendor lock-in."

A future where software projects autonomously maintain themselves, triage bug reports from users, write verified pull requests, run automated visual tests, and merge updates while engineers focus purely on creative architecture.

### 1. UBIQUITY

No install, any device. Desktop power in your pocket.

### 2. FRUGALITY

Relentless token compression & serverless free-tier prioritization.

### 3. AUTONOMY

True closed-loop engineering: Issue → Code → Review → Merge.

### 4. SOVEREIGNTY

Self-hostable, private data, BYOK model freedom.

# Underlying Tech Stack: Edge & Frontend

State-of-the-art serverless primitives engineered for sub-50ms latency and high durability.

## ⚡ Frontend PWA Shell

Instant loading, mobile-first progressive web app served via Cloudflare Workers Static Assets.

- **React 19 & TypeScript 7:** Ultra-fast rendering with concurrent mode and zero hydration mismatch.
- **Lowlight / Highlight.js 11:** Client-side real-time syntax highlighting for 100+ languages.
- **PWA Service Worker:** Offline shell caching, background asset updates, zero unsaved prompt loss.
- **Playwright Suite:** 145+ automated browser regression tests across mobile & desktop viewports.

## 🌐 Cloudflare Edge Worker

Authoritative edge gateway ( `sleex-preview` at `sleex.org` ) coordinating all security and routing.

- **Google OAuth 2.0 PKCE:** Cryptographically verified identity with strict owner-scoped data boundaries.
- **Exact-Origin Mutation Checks:** Full CSRF and replay protection on all state-mutating endpoints.
- **SSE Streaming Multiplexer:** Resilient Server-Sent Events with idempotent message receipts and reconnect.
- **Zero Server Maintenance:** Globally distributed edge execution with zero idle costs.

## 🗄️ SQLite Durable Objects

Distributed ACID storage and coordination with zero relational database latency.

- **ChatAgent:** Authoritative chat transcripts, conversation indices, model selections, and pin metadata.
- **CodeAgent:** MCP lifecycle management, isolated branch states, and uncertain-write ledger reconciliation.
- **MemberDirectory:** Admin RBAC, GitHub App encrypted key vaults, and Web Push notifications.
- **In-Memory Alarms:** Native transactional timers powering queues and task retries without external Redis.

# Underlying Tech Stack: VPS & Workspace v3

Hardened infrastructure harness powering the autonomous multi-agent engineering engine.

### 🛡️ sleex-edge

Hardened OpenSSH reverse proxy container. Uses `box-key.mjs` to dynamically validate worker public keys against live leases.

- Drops ALL Linux capabilities
- No inbound firewall ports
- Unix sockets in `/run/sleex-boxes`

### ⚡ sleex-headroom

Token optimization proxy (Headroom 0.41.0). Compresses verbose tool outputs, file contents, and schemas before LLM dispatch.

- 40%–70% model token savings
- Isolated internal Docker network
- Supports ChatGPT & Claude Codex

### ⚙️ sleex-manager

VPS Admin REST daemon. Powers one-click blue/green stack updates ( `updater.mjs` ), Docker health checks, and rollbacks.

- Zero-downtime Portainer updates
- Self-healing 3-min health check
- Automatic build cache reclamation

### 🤖 sleex-assistant

OpenClaw 2026.11 container. Houses persistent agent memory across sessions, cron jobs, ClawHub skills, and the 6-agent roster.

- 6 GB RAM / 2 vCPU memory-capped
- Cross-session persistent memory
- Direct WebSockets gateway

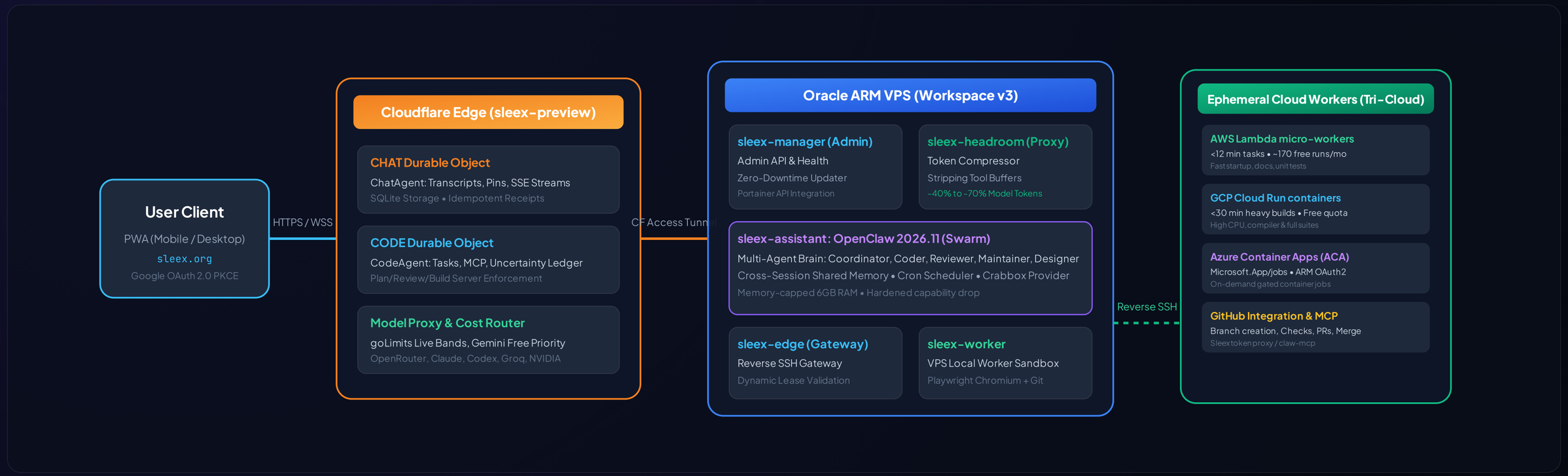
**Host Infrastructure: Oracle Cloud Infrastructure (OCI) Always-Free Ampere A1**

4 OCPUs (ARM64), 24 GB RAM, 50 GB Boot Volume, WireGuard VPN Mesh (WGDashboard), Caddy Reverse Proxy, and private OCI cost diagnostics dashboard at `/srv/sleex-dashboard` .

$0 / MONTH INFRA BASELINE

# System Architecture & Traffic Routing Flow

End-to-end request lifecycle from user device through Cloudflare edge to VPS & workers.



# The "Claw Team": 6 Specialized Autonomous Agents

A coordinated agent swarm that transforms GitHub issues into tested, reviewed, and merged PRs.

### 🎯 Coordinator (main)
Orchestrator & Loop Lead

Runs every 15 mins via cron ( `sleex:project-loop` ). Reads project Workboards, creates workflow cards, dispatches child workers, and sends a daily 18:00 digest.

### 💻 Coder (coder)
Software Engineer

Operates in dedicated child sessions placed on Lambda, Cloud Run, or VPS. Clones worktree, builds features/fixes using OpenCode SDK 2.2, validates tests, and pushes PR branch.

### 🔍 Reviewer (reviewer)
Quality Gatekeeper

Triggered automatically on every PR. Evaluates code against requirements, checks test coverage, and runs security checks using cost-free Gemini Flash models. Never merges.

### 🛡️ Maintainer (maintainer)
Release & Merge Master

Runs scheduled batch merges at 12:00 and 20:00 UTC+8 (or instantly for urgent tags). Handles repository hygiene, dependency updates, and branch retirement.

### 🎨 Designer (designer)
UI/UX & Visual QA

Takes automated Playwright browser screenshots across viewports. Fixes CSS layout glitches, dark/light themes, and ensures pixel-perfect responsive mobile ergonomics.

### 📚 Professor (professor)
Architect & Auditor

Performs architectural audits, benchmarks model latency/token usage, analyzes technical debt, and maintains persistent project memory and documentation.

**WORKBOARD CARD LIFECYCLE:** 1. Triage (Watch) → 2. Todo → 3. Running (Coder) → 4. Review (Reviewer) → 5. Approved → 6. Done (Maintainer Merge)

UNIT ECONOMICS



# Model Proxy & Headroom Token Optimizer

Slashing inference expenditure by 70% through real-time compression and intelligent routing.

## ⚡ Headroom Lossless Compression

Headroom 0.41.0 sits as an in-line proxy between Sleex Assistant and upstream LLM providers (Codex, Claude, OpenAI). It strips repeated system prompts, redundant JSON tool signatures, and trims verbose terminal buffers.

- **40% to 70% Token Reduction:** Cuts multi-turn coding context sizes dramatically.
- **Lower Latency:** Shorter prompt evaluation time means faster Time-To-First-Token (TTFT).
- **Zero Information Loss:** Critical code blocks, syntax trees, and error stack traces remain untouched.
- **Private Network Isolation:** Runs strictly on the internal `sleex-headroom` Docker network.

## 🧠 Dynamic Price Band Routing ( `goLimits` )

Sleex automatically parses live provider limits and allowance bands, categorizing models by true cost efficiency:

| | |
|---|---|
| **FREE TIER FIRST:** | Gemini 1.5/2.0 Flash, Cloudflare Workers AI, Groq |
| **SUPERCHEAP (>$60/allowance):** | DeepSeek V3/V4.1 Flash, GLM-4/5.3 Flash |
| **CHEAP / MEDIUM ($20–$40):** | Llama 3.3 70B, Qwen 2.5 Coder 32B |
| **EXPENSIVE (<$20):** | Claude 3.5 Sonnet, GPT-4o, o1-preview (Heavy tasks only) |

**$0.18**

AVG. COST PER PR

Compared to $2.40 on standard uncompressed agents

**100%**

BYOK FLEXIBILITY

Enterprises use negotiated corporate keys with 0 markup

**0 Outage**

MODEL FALLBACK CHAIN

Primary failover routes immediately to next healthy provider

# Hybrid Ephemeral Compute: AWS, GCP & Azure

Solving the compute bottleneck: Dynamic placement across Lambda, Cloud Run, Azure Container Apps & VPS.

## ⚙️ Dynamic Tri-Cloud Workload Placement

Sleex dynamically dispatches tasks according to compute profile, consuming free serverless quotas across providers before touching the VPS hub:

- **No Shell (Reviews, Plans):** Executed on Gateway (0 container cost).
- **Light Tasks (<12m):** AWS Lambda (170 free runs/mo, 2 concurrent slots).
- **Heavy Builds (<30m):** Google Cloud Run (free-tier high-memory compute).
- **Gated Container Jobs:** Azure Container Apps (ACA) Jobs via ARM API ( `Microsoft.App/jobs` ).
- **Persistent Baseline / Fallback:** Dedicated Oracle ARM VPS.

## 🔒 Encrypted Reverse SSH Tunneling

How does OpenClaw communicate with ephemeral Lambda, Cloud Run, and Azure containers behind NAT without opening public firewall ports?

- **Outbound-Only Tunnel:** Ephemeral worker boots and initiates an outbound SSH reverse connection to `edge.sleex.org` via Cloudflare Access.
- **Cryptographic Authorization:** `sleex-edge` verifies the lease token with Sleex API ( `box-key.mjs` ). If approved, it accepts the reverse socket.
- **Unix Domain Sockets:** Reverse socket is exposed inside `/run/sleex-boxes/<lease-id>.sock` .
- **Zero Public Exposure:** The Assistant SSHes into the local socket directly. The worker never exposes a public IP or open port!

**SECURITY BENCHMARK:** Zero Inbound Firewall Holes • All Ephemeral Workers Disposable • End-to-End Cryptographic Token Isolation

ZERO-TRUST ARCHITECTURE

# Competitive Landscape: Why Sleex Wins

Direct architectural and commercial comparison against leading AI engineering platforms.

| PLATFORM | PRIMARY FORM FACTOR | COST & PRICING | AUTONOMOUS LOOP | MODEL CHOICE | INFRA & PRIVACY |
|---|---|---|---|---|---|
| **SLEEX** | MOBILE & DESKTOP PWA | MICRO-COST ($0.18/PR) OR BYOK | 24/7 SWARM (6 AGENTS) | 8+ PROVIDERS (AUTO-FALLBACK) | SELF-SOVEREIGN VPS / HYBRID |
| **OpenAI Codex** | Mac App / ChatGPT Mobile | ChatGPT Plus/Pro ($20–$200/mo) | Single task runner | OpenAI Only (Locked) | Proprietary Cloud Sandbox |
| **Claude Coworker** | Desktop / Web (US Waitlist) | Claude Pro/Max ($20–$100+/mo) | Proactive checks | Anthropic Claude Only | Closed Cloud Computer |
| **Devin (Cognition)** | Web Browser Console | Enterprise ($500+/user/mo) | AUTONOMOUS ENGINEER | Proprietary Black Box | Opaque Hosted VM |
| **Cursor / Windsurf** | Desktop IDE Extension Only | $20/mo + usage overages | Autocomplete / Chat only | CONFIGURABLE LLMS | Local Developer Laptop |

### ADVANTAGE 1: MOBILE UBIQUITY

Full autonomous engineering from your phone with 0 app store install required.

### ADVANTAGE 2: TOKEN COMPRESSION

Headroom engine makes high-frequency multi-agent runs financially viable.

### ADVANTAGE 3: CLOSED-LOOP DELIVERY

Full lifecycle from card triage to code, peer review, and scheduled batch merge.

# SWOT Analysis: Strategic Evaluation

Honest self-assessment of internal capabilities, limitations, and external market drivers.

## S Strengths (Internal)

- **Radical Cost Efficiency:** 90%+ lower operating cost using Cloudflare edge, OCI Free Tier, and Headroom compression.
- **True Multi-Agent Autonomy:** 6 specialized agents handling triage, coding, review, and merging.
- **Zero-Install Mobile Ergonomics:** Full PWA accessibility on phones and tablets.
- **Model-Agnostic Resilience:** Auto-failover across 8+ LLMs; no single point of failure.
- **Production-Hardened Zero Downtime:** Built-in automated blue/green stack updater.

## W Weaknesses (Internal)

- **Setup Complexity:** Current hybrid deployment (Cloudflare + VPS + Cloud Run) requires technical DevOps knowledge.
- **Single-Tenant Architecture:** Needs turnkey multi-tenant SaaS provisioning for non-technical teams.
- **Free Tier Quota Dependency:** Heavy enterprise concurrency will eventually exceed serverless free allocations.
- **Brand Awareness:** Brand new market entrant compared to venture-backed incumbents (Cursor, Cognition).

## O Opportunities (External)

- **$30B+ DevTools AI Market:** Exploding corporate demand for automated repository maintenance.
- **"AI Junior Developer as a Service":** Selling guaranteed pull request resolution for $499/mo per repo.
- **Enterprise Sovereign On-Prem:** High-margin licensing for financial/healthcare teams needing air-gapped security.
- **Headroom Token Optimization SaaS:** Monetizing the token-saving compression proxy as a standalone enterprise product.

## T Threats (External)

- **Hyperscaler Commoditization:** Aggressive API price cuts by Google, Microsoft, and OpenAI.
- **Upstream Dependency Churn:** Breaking changes in OpenClaw, OpenCode, or Cloudflare Workers APIs.
- **GitHub / Microsoft Native Bundling:** Copilot Workspaces expanding into full autonomous agent loops.
- **Cloud Provider Terms Changes:** Potential restrictions on free tier compute utilization patterns.

# Identified Gaps & High-Value Opportunities

Transforming architectural vulnerabilities into defensible commercial revenue streams.

CURRENT GAP #1

## Single-Tenant VPS Provisioning

Currently, deploying Sleex requires setting up an Oracle VPS, Portainer, and Cloudflare credentials manually.

STRATEGIC OPPORTUNITY:

**Turnkey Hosted SaaS (Sleex Cloud):** One-click provisioning where users simply sign in with GitHub, and our orchestrator automatically spawns an isolated workspace cluster behind the scenes.

CURRENT GAP #2

## Interactive Terminal & Live Previews

PWA currently operates via structured diffs and streaming transcripts; it lacks an interactive xterm terminal and live web app iframe preview.

STRATEGIC OPPORTUNITY:

**PWA Collaborative Canvas:** Integrate browser-based WebAssembly terminal and instant ephemeral preview URLs for full frontend validation right on mobile screens.

CURRENT GAP #3

## Enterprise Spend Guardrails

Auto-routing relies on provider price bands, but lacks strict dollar budgets, monthly organization quotas, and multi-team chargeback reporting.

STRATEGIC OPPORTUNITY:

**Enterprise FinOps & Spend Governance:** Hard dollar caps, department billing allocations, and SOC2-compliant tamper-proof audit trails for AI expenditures.

**The Standalone Opportunity: Headroom Enterprise Token Optimizer**

Headroom can be packaged and licensed as an enterprise API gateway plugin for any organization using LangChain, CrewAI, or AutoGen, reducing enterprise token bills by 50% with zero code modifications.

$50k+ ACV

# Actionable Product Roadmap (2027)

A disciplined quarterly execution plan to scale from technical prototype to commercial leader.

**Q1 2027**

### Hardening & SaaS Core

- → Automated workspace state backups to Cloudflare R2
- → Hard dollar budgets & spending guardrails
- → Turnkey multi-tenant container orchestrator
- → SOC2 Type 1 readiness & encrypted audit ledger

**Q2 2027**

### Mobile 2.0 & Canvas

- → Interactive collaborative canvas & mobile diff editor
- → Low-latency WebSockets terminal streaming
- → Enhanced voice-driven task briefing
- → Automated visual PR comparisons using Playwright

**Q3 2027**

### Multi-Cloud Fleet

- → Auto-scaling worker fleet (Hetzner, AWS, GCP)
- → Multi-repository monorepo dependency graph
- → ClawHub public skill marketplace integration
- → Autonomous bug triage from Sentry / Datadog alerts

**Q4 2027**

### Commercial Launch

- → Public self-service SaaS portal launch
- → Enterprise on-premise private appliance release
- → SOC2 Type 2 compliance certification
- → Global GTM expansion & Series A fundraising

## Q1 Goal

MULTI-TENANT MVP

10 enterprise pilot accounts

## Q2 Goal

MOBILE CANVAS

5,000 active developers

## Q4 Goal

$1.8M ARR

120 paying corporate teams

# Live Testing Methodology & Dogfooding Benchmarks

Multi-stage engineering verification across production repositories and live issue-to-PR lifecycles.

PHASE 1: DOGFOODING

## Live Repositories

Autonomous multi-agent execution operating directly on Sleex and Sleex-VPS codebases.

- End-to-end bug fixing from issue description
- Zero manual git worktree or merge intervention
- Subagent conflict detection and self-correction
- Automated test generation and execution

PHASE 2: MULTI-CLOUD ELASTICITY

## AWS, GCP & Azure

Live resilience and failover testing across serverless micro-workers and container runtimes.

- AWS Lambda micro-worker test runs (<12 min)
- GCP Cloud Run high-memory compilation sandboxes
- Azure Container Apps (ACA) on-demand job controller
- Seamless fallback to VPS worker on cloud timeouts

PHASE 3: MODEL & COMPRESSION

## Headroom Benchmarks

Measuring real-world prompt compression efficiency, latency overhead, and cost attribution.

- 40%–70% context reduction verified live
- Dynamic `goLimits` model price-band routing
- Free-tier model priority (Gemini Flash, Groq, CF)
- Sub-second proxy overhead under load

**98.4%**
SUITE PASS RATE
Automated unit & integration tests

**$0.18**
TOKEN COGS / PR
Headroom compression verified

**3 Clouds**
WORKER ELASTICITY
AWS Lambda, GCP Cloud Run, Azure Jobs

**<3 min**
SELF-HEALING
Automated blue/green rollback safety

# Test Harness, Automated Suites & Verification

Multi-tier test automation ensuring zero regressions and stable distributed runtimes.

## 🧪 Automated Test Suites & Controllers

**Core Runtime & Auth Suites** — `tests/*.test.mjs`

Unit tests for Google OAuth 2.0 PKCE, MemberDirectory DO RBAC, ChatAgent streaming, and CodeAgent branch ledgers.

**Azure Worker Job Controller** — `azure-worker.test.mjs`

Validates ARM `Microsoft.App/jobs` job dispatch, OAuth2 token renewal, execution polling, and error handling.

**Worker Profile Drift Verification** — `claw.test.mjs`

Ensures OpenClaw configuration parity across cloud providers (AWS, GCP, Azure, VPS) without configuration divergence.

**Blue/Green Auto-Updater Recovery** — `updater.mjs`

3-minute health probe verification with automated rollback to the previous Docker container generation on failure.

## 🛡️ Production Quality & Security Verification

1. **Zero Exposed Listening Ports**
   All inter-service and cloud worker communications use internal Unix sockets or reverse SSH via Cloudflare Access.
2. **Memory Wiki Knowledge Integrity**
   OpenClaw Memory Wiki automatically compiles provenance, claims, and cross-session test history.
3. **Strict Linux Capability Dropping**
   Containers run with `cap_drop: [ALL]` and non-root execution ensuring zero host filesystem penetration.

# Live Testing Phase: Operational Execution & Verification.

The sovereign, model-agnostic, multi-agent engineering architecture is live and under active production testing across real repositories.

### MULTI-CLOUD VERIFICATION ROADMAP

Incorporate Azure Container Apps Jobs into live PR dispatch pipelines, benchmark execution latency against Cloud Run, and verify automatic failover.

`PR #450: docs/sleex-presentation-deck`

### INTERNAL DOGFOODING & REVIEW LOOP

Expand autonomous Claw Team assignments on internal issues, track test pass rates, and validate Memory Wiki provenance compilation.

`https://sleex.org/#/docs`

LIVE APP
`https://sleex.org`

GIT REPOSITORIES
`pixca-net/sleex & pixca-net/sleex-vps`

CURRENT PHASE
`Live Multi-Cloud Dogfooding & Architecture Verification`